Problem – Solution Fit

Project Name: Smart Meet – Video Conferencing Application

◈ Customer Problem:

Users (students, professionals, teams) face difficulty in conducting smooth and efficient online meetings due to:

- Complex user interfaces
- Paid features in existing platforms
- Lack of integrated communication tools

◈ Customer Behavior:

- Uses multiple apps (Zoom, Google Meet, WhatsApp)
- Switches between platforms for chat, video, and scheduling
- Faces confusion and inefficiency

◈ Existing Solutions:

- Zoom (paid features)
- Google Meet (limited features)
- Microsoft Teams (complex UI)

◈ Gaps Identified:

- No single simple platform with all features
- High dependency on multiple tools
- Poor user experience for beginners

◈ Proposed Solution Fit:

Smart Meet provides:

- Simple UI
- Free video conferencing

- Integrated chat + video + meeting system
- Easy room creation and joining

◈ **Value Proposition:**

- One platform for all communication needs
- Easy to use
- Cost-effective solution
- Real-time collaboration

◈ **Outcome:**

Users can:

- Join meetings easily
- Communicate efficiently
- Collaborate smoothly without switching apps

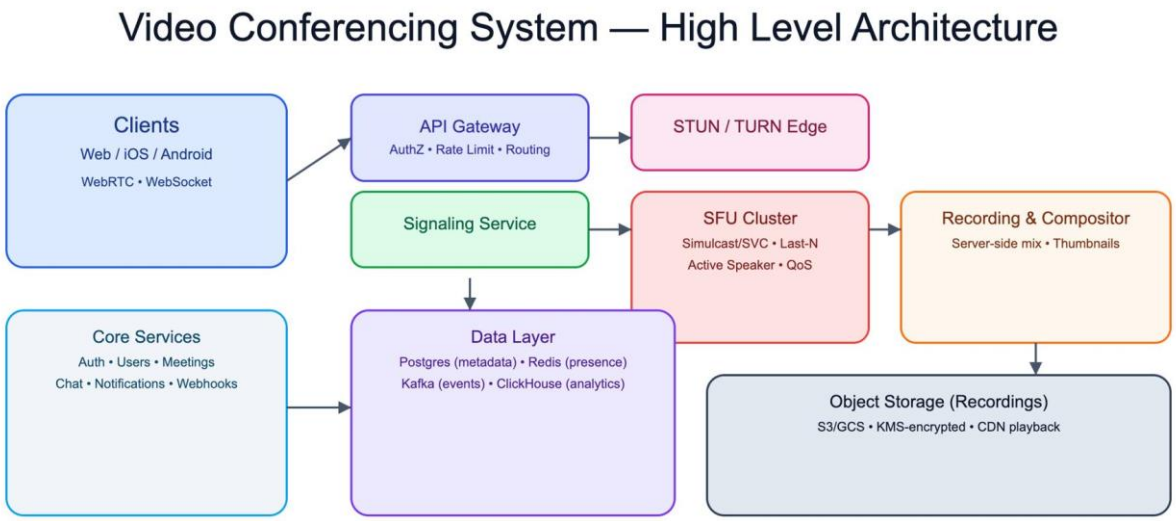
Proposed Solution

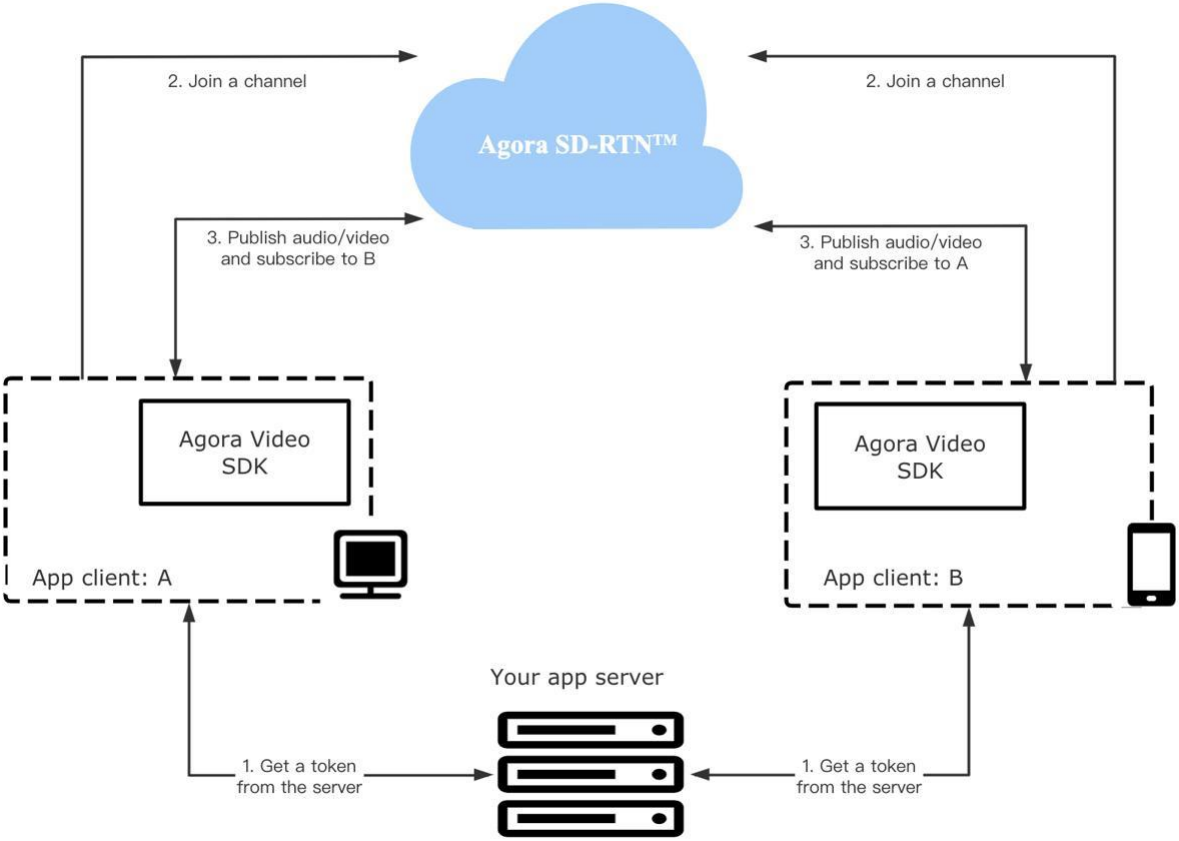
Project Name: Smart Meet – Video Conferencing Application

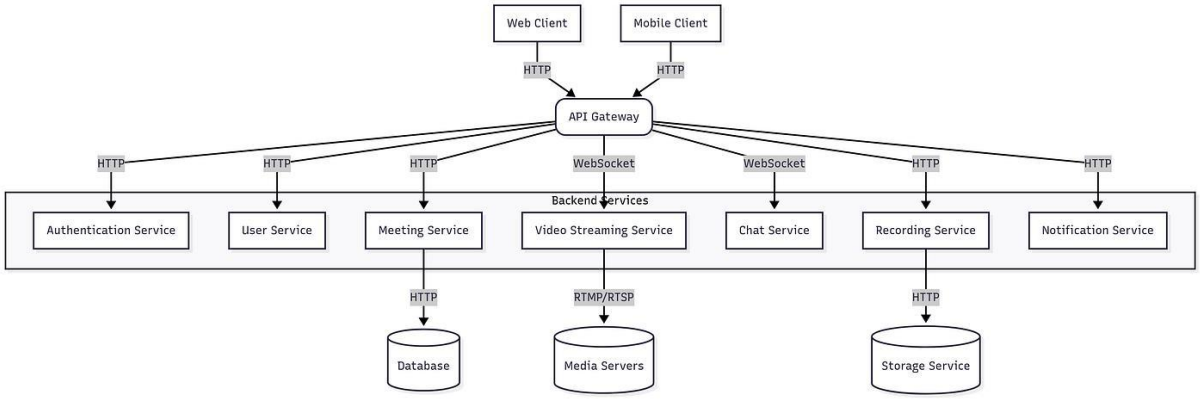
S.No	Parameter	Description
1	Problem Statement	Difficulty in managing online meetings due to complex tools and lack of integration
2	Idea / Solution	A web-based video conferencing app using MERN stack, Socket.io, and Agora
3	Novelty / Uniqueness	Combines video, chat, and meeting features in a simple interface
4	Social Impact	Helps students, teachers, and teams collaborate easily
5	Business Model	Freemium model (basic free, advanced paid features)
6	Scalability	Can support multiple users and expand to mobile apps

SOLUTION ARCHITECTURE

◈ Architecture Diagram







📝 Architecture Explanation

◈ Solution Architecture Description

The Smart Meet system follows a 3-tier architecture:

□ 1. Presentation Layer (Frontend)

- Built using React.js
- Handles user interface and interactions
- Features:
    - Login/Register

    - Create/Join meeting
    - Chat & controls

⚙️ 2. Application Layer (Backend)

- Built using Node.js + Express.js
- Handles:
    - API requests
    - Authentication (JWT)
    - Room management
    - Socket.io communication

🔄 3. Real-Time Communication Layer

- Socket.io → Chat & signaling
- Agora RTC → Video/audio streaming

🗄️ 4. Data Layer

- MongoDB
- Stores:
    - User data
    - Meeting details

🔗 5. External Services

- Agora API
    - Video streaming
    - Media handling

🔁 Data Flow:

1. User logs in → request sent to backend
2. Backend verifies → response returned
3. User joins room → socket connection created
4. Agora handles video streaming
5. Messages sent via Socket.io

✅ Key Features of Architecture:

- Real-time communication
- Scalable design
- Modular components
- Secure authentication